SLIIT
FACULTY OF COMPUTING

FINAL PRESENTATION (PP3) · R26-IT-143

AI-Based Smart Swimming Pool Monitoring and Maintenance System

Four AI modules · one live dashboard · Centre of Excellence in AI (CEAI) · BSc (Hons) IT



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PP2 FEEDBACK

PP2 Comments and Actions Taken

Member	Panel comment at PP2	Action taken
IT22156310 Crowd	No idea about the bather load added into the interface.	Bather load explained: person-hours of swimming today (swimmers × time in water). Each action has a threshold, e.g. chlorine dose at 20, backwash at 50, with Medium / High / Critical priority.
IT22247254 Water	Explain how the output is rated Critical (value ranges).	Critical ranges shown with each result: turbidity > 5 NTU or TDS ≥ 1500 ppm; pH outside 6.8–8.0 is unsafe. Every Warning / Critical states its reason.
IT22248480 Drowning	Drowning count not accurate on the videos.	New model integrated. Alert only when ≥ 60% of recent frames agree, one alert per incident (not per frame); conflicting labels on one swimmer filtered.
IT22277336 Garbage	Unclear what an “event” is; detect tiny and submerged garbage too.	“Events” renamed “Incidents” with an (i) explanation and a snapshot feed. Added submerged-garbage detection with surface / under-water counts.

PROBLEM IN BRIEF

Pool safety and hygiene still run on manual, disconnected checks

Safety

Drowning is often silent. Lifeguard attention drops over long shifts, and swimmers hide each other in busy water.

Water quality

Chemistry is tested by hand a few times a day. Water can turn unsafe between tests without anyone knowing.

Maintenance

Cleaning follows a fixed calendar, not real usage, so effort is wasted on quiet days and late on busy ones.

Hygiene

Litter on the surface or sunk to the floor is noticed late, harming hygiene and filters.

The core gap: these four signals are never reasoned about together, so every decision is reactive.

DESIGN EXCELLENCE

How our solution goes beyond existing systems

Capability	Typical existing solutions	Our system
Scope	One function per product	Four AI modules in one platform
Maintenance	Fixed calendar	Triggered by measured bather load
Water quality	Manual test kits, pH-only control	Live sensors + ML class, with reasons
Drowning	Costly underwater camera systems	One standard camera, stable alerts
Garbage	Manual inspection	Surface + submerged; balls never alert
Decision layer	None: staff combine signals	Decision engine fuses all modules
Deployment	Fixed, expensive hardware	Existing camera + low-cost sensor node

OVERALL SOLUTION

One camera, one sensor node, one decision view

1 · Capture
Pool camera + Arduino water sensors

→

2 · AI inference
Crowd, drowning, garbage, water models

→

3 · Events
Bather load, persistence, incidents

→

4 · Decision
Risk level and maintenance urgency

→

5 · Dashboard
Live alerts, reports, plans





Tested on site

Deployed live at a public swimming pool with real swimmers, phone camera feed and the poolside dashboard.

COMPONENT 1 · INDIVIDUAL PART

Crowd-Aware Maintenance Scheduling

Counts swimmers in the water from the shared camera and turns real pool usage into maintenance actions.

Swimmer detection

YOLO11m person detector, chosen after it found 9 people where our custom model found 1

In-water filter

Pool-region polygon, so people on the deck never inflate the count

Bather load

Person-hours of swimming today; each action fires at its own threshold

Integration lead

Shared camera, vision scheduler, decision engine, plans and billing



0.794

mAP@50, custom swimmer model (test split)

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COMPONENT 2 · INDIVIDUAL PART

AI-Based Water Quality Prediction

Reads live values from an Arduino sensor node and classifies the water as Safe, Warning or Critical, with the reason.

Parameter	Safe range	Critical when
Turbidity	≤ 1 NTU	> 5 NTU
TDS	≤ 1000 ppm	≥ 1500 ppm
pH	7.2 – 7.8	< 6.8 or > 8.0
Temperature	Logged with each reading	Context only

XGBoost classifier (5 inputs, 3 classes) · 60-second averaged final reading · alert only when the state changes · simulation mode for demos



3 classes

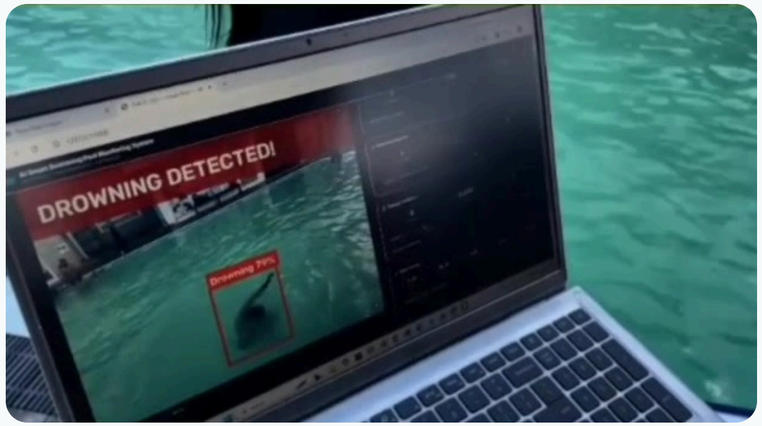
Safe · Warning · Critical, each with its reason

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COMPONENT 3 · INDIVIDUAL PART

Swimmer Behaviour and Drowning Detection

Classifies each swimmer as Swimming, Drowning or Out of water and raises one stable alert per incident.



3-class YOLOv8s model at 640 px input

Sliding window: danger at ≥ 60% of recent frames, clears at ≤ 20%

One alert per incident, not one per frame

Conflict filter for two labels on one swimmer

Above: “Drowning detected” raised live during the public-pool field test.



0.907
mAP@50

0.883
Recall

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COMPONENT 4 · INDIVIDUAL PART

Garbage Intrusion Detection

Finds litter on the surface and under the water, explains each incident in plain language, and never alerts on balls or leaves.

Surface and submerged

Separate counts for floating litter and garbage under the water

New object on floor

Flags something new that stays still on the pool floor

Incidents, explained

One confirmed event = one incident, with snapshot, place and an (i) guide

Who dropped it

Person overlay and pose cues describe the likely action behind the litter



0.949
mAP@50

0.927
Recall

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RESULTS AND TESTIMONIALS

Measured accuracy and a live field test

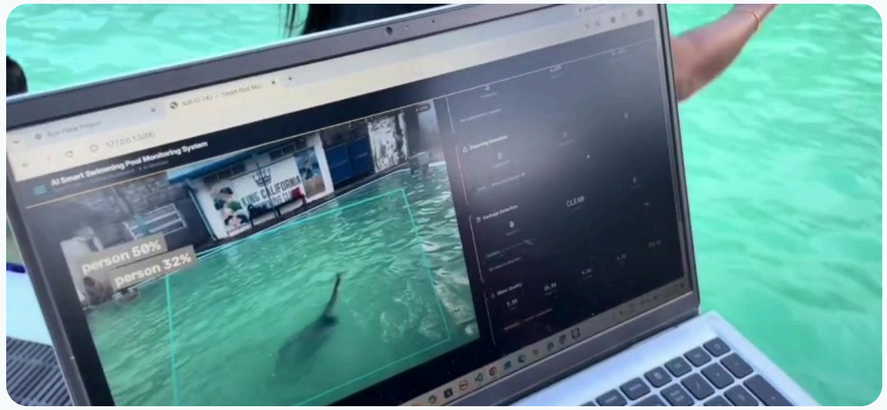
Module	Precision	Recall	mAP@50	mAP@50-95
Drowning	0.877	0.883	0.907	0.594
Garbage	0.932	0.927	0.949	0.446
Crowd (custom)	0.777	0.760	0.794	0.338
Water (XGBoost)	1.00	1.00	–	–

Held-out test / validation splits. Water scores are on a synthetic, perfectly separable dataset, so they describe the data, not field accuracy.

4 / 4
modules integrated

27 / 27
system checks pass

18.3
FPS on CPU



USER FEEDBACK

"[Add the pool manager's or lifeguard's comment from the field test]"

— [Name, role, pool]

LIMITATIONS AND FUTURE IMPROVEMENTS

What we would build next

Limitations

- Drowning training data is staged, not real incidents
- Water model trained on synthetic data; pH probe needs recalibration
- One camera angle: occlusion in crowded water
- Tested at a single pool site
- Phone camera apps allow only one viewer at a time

Future improvements

- Retrain the water model on logged real sensor data
- Add underwater and multi-angle cameras
- SMS and mobile push alerts for lifeguards
- Edge device deployment (e.g. Jetson) at the pool
- Multi-site validation with partner pools

R26-IT-143 · FINAL PRESENTATION

Thank you

Questions, then the live demonstration

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